

Class 8 Sets

Goals for the class:

1. Become familiar with notation and terminology for sets.
2. Understand the connections between set theory and logic.
3. Know strategies to prove two sets are equal.

Notation

Algebra of sets

The algebra of sets is exactly parallel to the algebra of sentential logic. It is another example of a Boolean algebra.

Class Activity Write a set theory expression that corresponds to a line in the table below.

Formal Name	Logical Equivalence
Double negation	$\sim (\sim p) \equiv p$
De Morgan's laws	$\sim (p \wedge q) \equiv (\sim p) \vee (\sim q)$ $\sim (p \vee q) \equiv (\sim p) \wedge (\sim q)$
Implication expansion Contrapositive Biconditional expansion	$p \rightarrow q \equiv (\sim p) \vee q$ $p \rightarrow q \equiv (\sim q) \rightarrow (\sim p)$ $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
Commutativity	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associativity	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributivity	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Tautology Contradiction	$p \vee (\sim p) \equiv T$ $p \wedge (\sim p) \equiv F$
Simplification	$p \wedge T \equiv p$ and $p \wedge F \equiv F$ $p \vee T \equiv T$ and $p \vee F \equiv p$

Proof Strategy: How to prove two sets are equal.

$A = B$ means every element of A is in B and every element of B is in A .

Example: Prove $(A \cup B)^c = A^c \cap B^c$

C

Disprove the statement: *If $A \not\subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.*

Other ways to create new sets from given sets besides intersection, union, and complement.

Difference of two sets

Cartesian Product

Power set

Examples: Let $A = \{1,3,5,7,9\}$, $B = \{2,3,5,7\}$ and $C = \{1,2,3\}$.